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Experiment Name: Z test, T test, F Test, Anova Test T - Test

A sample of 10 students scored the following marks in an exam: [72, 68, 75, 70, 74, 69, 71, 73, 70, 72]

We want to test whether the average mark = 70 (μ0 = 70) at 5% significance level using python

```
import numpy as np from scipy
import stats
marks = np.array([72, 68, 75, 70, 74, 69, 71, 73, 70, 72]) mu_0
= 70
t_stat, p_value = stats.ttest_1samp(marks, mu_0) print(f"Tstatistic:
{t_stat:.3f}") print(f"P-value: {p_value:.4f}") alpha = 0.05 if
p_value < alpha:     print("Reject Null Hypothesis → Mean is
significantly different from 70.") else:     print("Fail to Reject
Null Hypothesis → No significant difference.")
```

```
T-statistic: 1.993
P-value: 0.0774

Fail to Reject Null Hypothesis → No significant difference.
```

Z - Test

```
Z-statistic: 2.400
P-value: 0.0164

Reject Null Hypothesis → Mean is significantly different from 50 g.
```

Anova Test

```
import numpy as np from math
import sqrt from scipy.stats
import norm x_bar = 51.2 mu_0
= 50 sigma = 3 n = 36
z_stat = (x_bar - mu_0) / (sigma / sqrt(n)) p_value = 2 * (1 -
norm.cdf(abs(z_stat))) print(f"Z-statistic: {z_stat:.3f}") print(f"P-
value: {p_value:.4f}") alpha = 0.05 if p_value < alpha:
print("Reject Null Hypothesis → Mean is significantly different
from 50 g.") else:     print("Fail to Reject Null Hypothesis → No
significant difference.")
```

Three fertilizers (A, B, C) were tested on crop yield (in kg). Is there a significant difference among fertilizers? (Use α = 0.05)

Fertilizer Yields

A 20, 22, 23

B 19, 20, 18

C 25, 27, 26

```
import numpy as np from scipy
import stats
A=[20,22,23]
B=[19,20,18] C=[25,27,26] f_stat,p_value=stats.f_oneway(A,B,C)
print(f"F-statistic: {f_stat:.3f}") print(f"P-value: {p_value:.4f}")
alpha=0.05 if p_value<alpha:     print("Reject Null Hypothesis-Means
are significantly different.") else:     print("Fail to Reject Null
Hypo
```